

Game design Doc
for
One man planet (temp)

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Version: V1.14

Please keep the version number updated whenever you update the document. Nothing is finalized and this meant to be the start of our conversation, please feel free to jump in and contribute.

1/2/2023	Jeep	created the doc	v 1.0
1/3/2023	Jeep	added screen size	v 1.1
1/4/2023	Jeep	updated storyline, sprite list	v1.2
1/5/2023	Jeep	changed to sci-fi lore, crafting mechanic, boss events, project plan	v1.3
1/7/2023	Jeep	splitting prototype/2nd/3rd deliveries	v1.4
1/8/2023	Jeep	added player, monsters ability, updated core loop	v1.5
1/8/2023	Jeep	added table of content	v1.6
1/10/2023	Mario	added suggestion under Features/mechanics	v1.7
1/11/23	Andrew	QC pass	v.1.8
1/12/2023	Jeep	updated project plan, moved some tasks to post prototype	v1.9
1/12/2023	Andrew	formatting, introduction, change phase 3 color to green for visual clarity, change prototype color to red for urgency	v1.10
1/13/2023	Anthony	revised writing and formatting	v1.11

1/14/2023	Jeep	added tech tree, gametech dependencies	v1.12
1/14/23	Andrew	Editing, add diagrams, monster behavior	v1.13
1/14/23	Anthony, Andrew, Jeep, Mario	Final QC pass before submission	v1.14

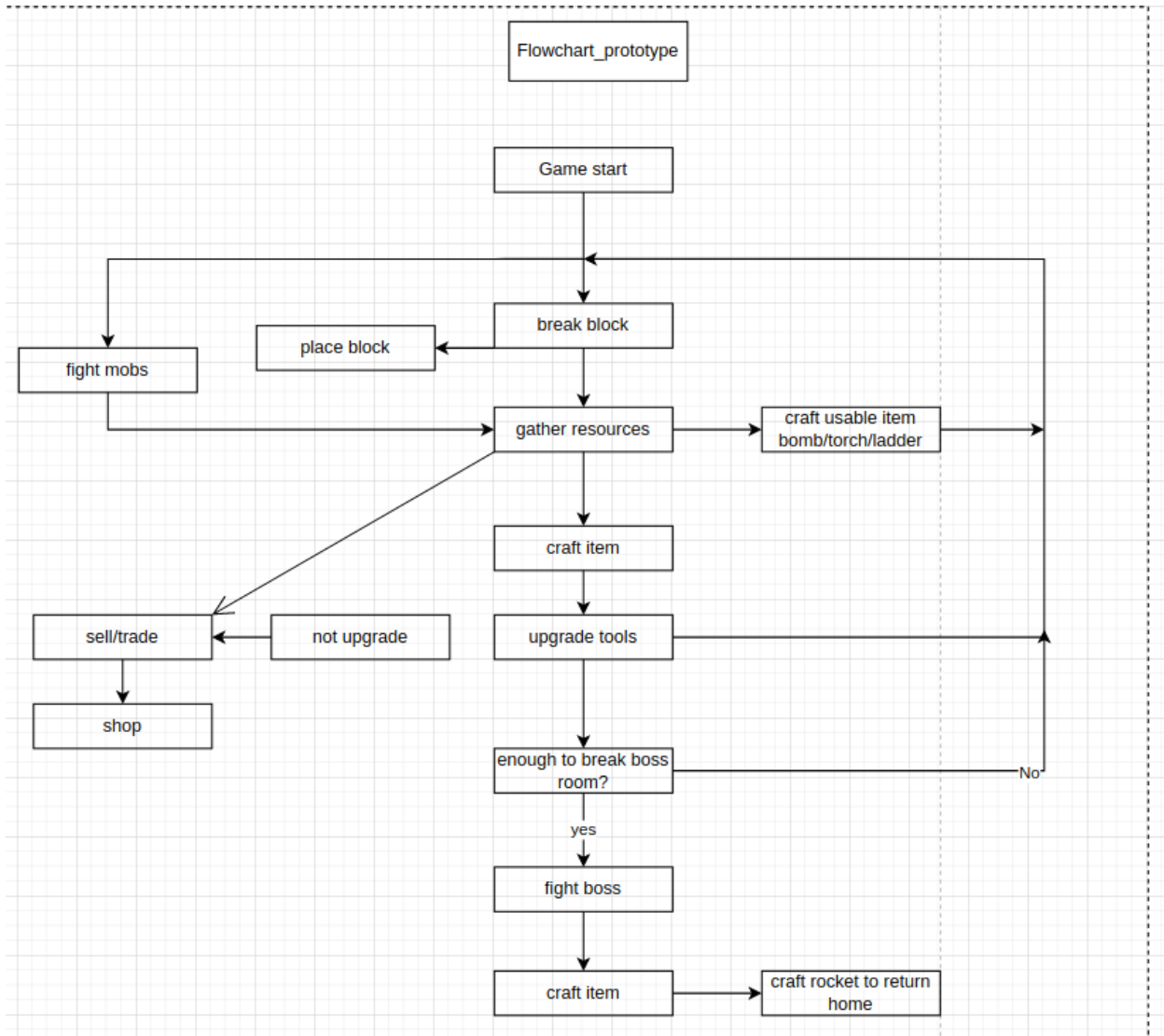
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Introduction

The goal of this project is to build a side-scrolling crafting survival game similar to Terraria. It will feature destroyable and placeable terrain, procedural generation, an extensive crafting system, several different monsters, and a handful of boss fights.

Gameplay

The player digs the terrain and fights various types of monsters to gather necessary resources to upgrade his harvesting tools and weapons in order to gather stronger blocks. There are items and plants that can be collected to boost his stats. The player also needs to unlock the necessary tech tree in order to build a ship to return home. (end goal)



Goal/Story

The player's avatar is a ship engineer and the sole survivor of a crash landing on Planet 2x359920. He wakes up (on the surface?) at the start of the game. The tutorial will then instruct the player to dig underground in order to stay safe. They must learn to avoid an ongoing meteorite bombardment of the surface by digging with the basic set of tools (short pulse laser, first tier).

To survive, players must explore the underground and create new weapons and space suits in order to navigate the harsh environment. They must fix the pulse drive and other necessary parts in order to space jump to a set of coordinates to be rescued by their employing corporation.

Once the player has killed fifty of the planet's minions, the first boss will spawn underground. After defeating it, the meteor shower will stop, allowing the player to safely (possibly?) explore the surface.

Subsequently, a new resource material will spawn in the world. The player can harvest this new mineral type to create better gear. They will need to go to the surface to use the character's communication tool for the first time which accidentally summons a new boss. This second boss will be a flying type since the meteorites are not a threat to it anymore. Once it is defeated, it will spit out a giant unbreakable egg on the surface.

New monster type(s), Broodlings, will appear in the world at random. These creatures can be defeated to obtain a necessary material to make a new crafting station, which will be essential to making super armors and weapons.

The final mission is to craft a photon gun to harvest the eggshell material which is necessary to deal with engine overheating when exiting the atmosphere. The act of successfully breaking the egg will spawn the final boss, the Broodmother (the story will lead to breaking the egg from npc/shop...TBD)

Features/mechanics

Scope for prototype in red

Scope for 2nd deliverable in orange

Scope for 3rd deliverable in green

- block breaking
 - 5 stages of sprite total, this will be serialized with numbers so higher power pickaxes will only take a single blow to break the lowest grade block (dirt)
 - need an ability to add sound to a certain entity that can/can't break
- item climb (need to finalize theme and story)
- Item durability (to encourage constant crafting)
- crafting mechanic (pending item climb and theme)
- random generate terrain
 - with caves (pending screen size and block size)
- inventory system
- UI design(search for Terraria screenshot)
- boss fight (at least one)
- block placing (putting block/material back into the world)
- animation system (run, walk, jump, fight, throw, shoot, death, idle, projectile)
- save/load/spawning mechanic (spawn point relocation option)
- game controller (1-9 for tool selection, aswd, mouse click, space bar)

- boss fight design,mechanic (optional pending goal choice)
- ECS design pattern (or anything better?)
- light/shadow (explore/unexplored area)
- mini map (optional)
- growing crops (also need day/night feature for the game) (optional)

Monster mechanics (in Characters section)

Characters

Player



- Health:
- Walk
- Jump
- Attack
 - Sword (melee)
 - Laser gun (ranged)

Monsters

Flying types

bloodsucker - fly fast straight to target to do damage (yields keratin)



- Class: Flying
- Health:

- Damage:
- Range: Melee
- Speed: Fast
- Behavior: Follow player
- Drops: **keratin**

mossFly - fly random slow (yield wood)



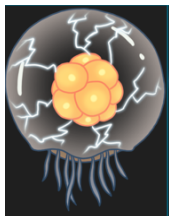
- Class: Flying
- Health:
- Damage:
- Range: Melee
- Speed: Slow
- Behavior: Random movement
- Drops: **wood**

wasp - slow movement- explode in contact



- Class: Flying
- Health:
- Damage:
- Range: Melee
- Speed: Slow
- Behavior:
 - Follows player
 - Explodes on contact
- Drops: **none**

lightJelly - stun when in contact



- Class: Flying
- Health:
- Damage: Stuns player on contact
- Range: Melee
- Speed:

- Behavior: Random movement
- Drops: **none**

lightBug - follow player around for 30 sec (act as free torch)



- Class: Flying
- Health:
- Damage:
- Range:
- Speed:
- Behavior:
 - Follows player for 30 seconds
 - Gives off light
- Drops: **none**

Crawlers

dirtCarver - slow but tough to kill



- Class: Crawling
- Health: High
- Damage:
- Range: Melee
- Speed: Slow
- Behavior:
- Drops: **none**

lightCrawler - no threat, but can be killed (critter)



- Class: Crawling
- Health:
- Damage:
- Range: Melee
- Speed:
- Behavior:
- Drops: **none**

wormTank - can only be killed by bomb, run toward the player (yield mineral, hide)



- Class: Crawling
- Health: Only killable with bomb
- Damage:
- Range: Melee
- Speed:
- Behavior: Follows player
- Drops: **mineral, hide**

creeperilla - live underground, run, shoot web ball (yields spider silk)



- Class: Crawling
- Health:
- Damage:
- Range: Ranged
- Speed: Fast
- Behavior:
- Drops: **spider silk**

Fixed spawn

grapeBomb - explode within 2 sec from contact (yields sticky stuff)



- Class: Fixed
- Health:
- Damage:
- Range: Melee
- Speed: Stationary
- Behavior: Explodes 2 seconds after contact
- Drops: **sticky stuff**

mossAmber - can be killed (yields wood)



- Class: Fixed
- Health:
- Damage:
- Range:
- Speed: Stationary
- Behavior:
- Drops: **wood**

spore - plant type (yields wood)



- Class: Fixed
- Health:
- Damage:
- Range:
- Speed: Stationary
- Behavior:
- Drops: **wood**

boss - TBD

Tech tree

Crafting

Crafting has 2 parts

1 craft without table (self craft) - ladder

2 craft with table/building - need to be within 2 block range to use the crafting menu

- the crafting menu will highlight if all requirement materials are in the bag
- if the item can't be crafted yet, the button will be grayed out

- recipe button will show all recipes for all items that can be craft from that station
- crafting stations include engineering hub, furnace, worktable, recycler, anvil
- the deep space antenna will only act as a mailbox (2 stages, will light up if has something new for the player to read)

Item diagram

for harvesting

- pickaxe

for combat (melee) - player will learn early that the harvest tool cannot kill the mobs

- laser sword (lightsaber type)

for combat (range)

- bow
- laser gun

armor (to keep it simple, there will only be a helmet)

- same as pickaxe with option to enchant

Craftable items

weapons (laser sword, projectile launcher, pickaxe)

ladder

torch

~~recycler (to discard item)~~

tradeStation (to buy/sell) a friendly alien will spawn inside, act as a shop

worktable (to upgrade craft item)

sleeping pod (to change spawn point and save game)

furnace (to make metal bars)

anvil (to craft alloy, combine 2 different bars)

toolbelt (to expand item slots +4 each) (optional)

engineering hub (to upgrade metal bar, coat with substance)

antenna (to get tips for playing the game, can be crafted with tier 1 material)

deep space antenna (to communicate with the corporation and summon the flying boss)

weapon melee/range (craftable at worktable, upgradable at engineering hub)

Harvestable plants (all effects are stackable) need to craft into a potion form.

firework - Hanenite nitrate (yield gunpowder to craft bomb)

brood saliva - Yuka Salivariae (sticky substance, combined with a regular bomb to make a sticky bomb)

MorningFin - Finomytozoa (health regen for a period of time)
 Batsnip - Nepeta Batoreil (temporary yield an ability to slow fall)
 LatteMelon - Cucumis Rubia (temporary yield an ability to speed mining combo)
 CakeWalk - Taurina Giggatoire (temporary yield an ability to move 15% faster)

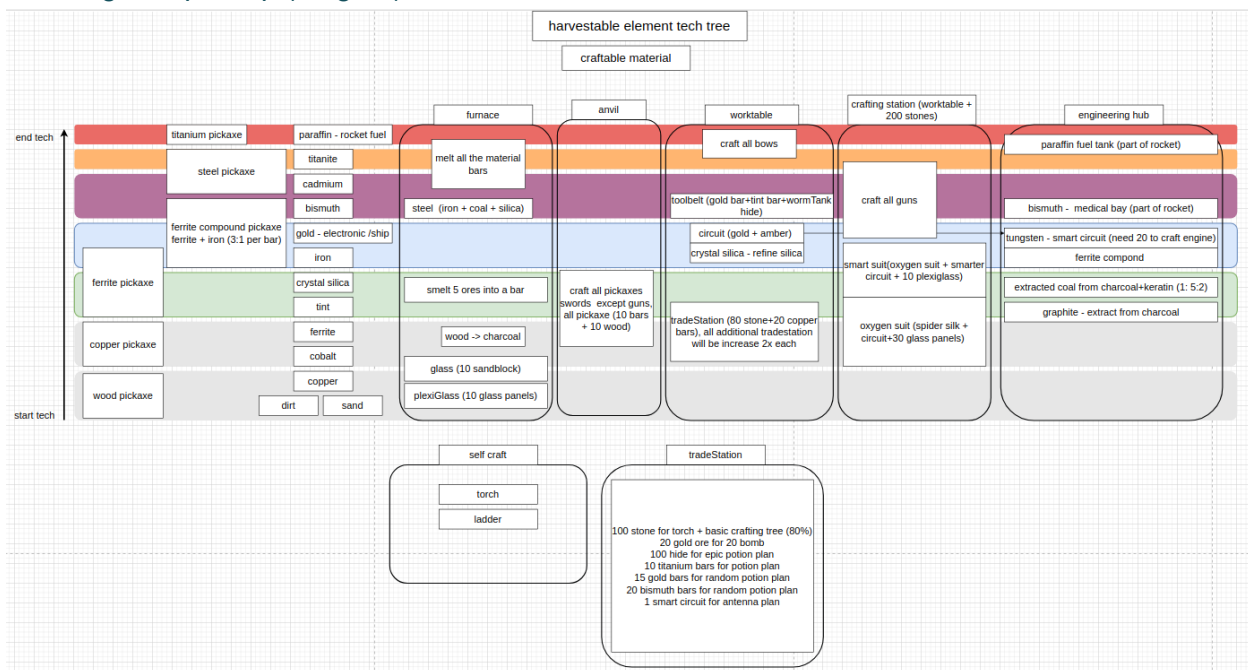
Weapon/armor-coating

item can be enchanted to get one of the followings

add life regen

add range for harvesting

add range for pick up (magnet)



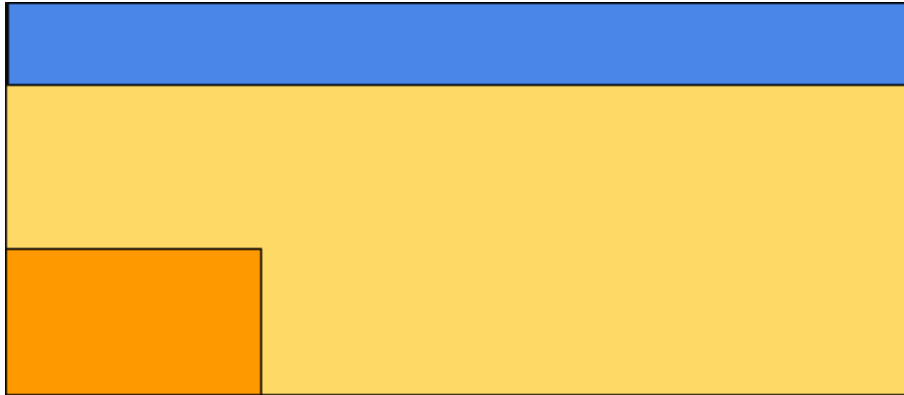
[Full Size](#)

Screen size/resolution

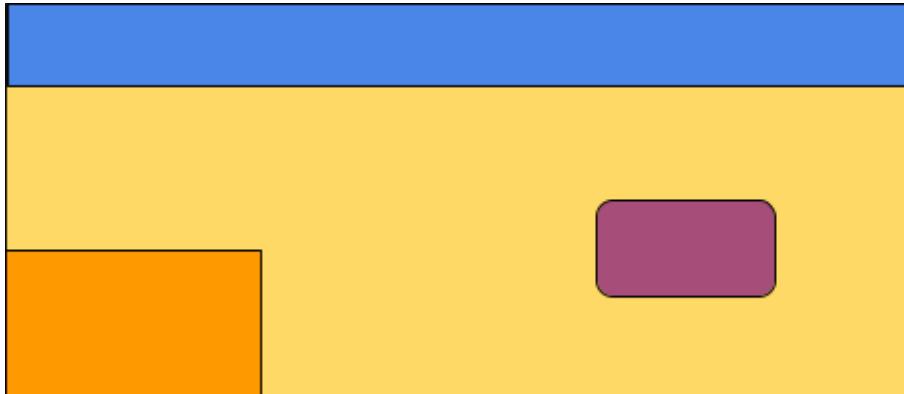
1024 x 768 px game screen (camera)

3200 x 1800 px map size (world)

Orange = screen size, Yellow = underground terrain, Blue = sky



Violet: 1st boss room size (randomly generate within the terrain) $\frac{2}{3}$ of game window resolution



Game object category

Visible

- material blocks
- harvestable plant
- plant material
- weapon (range, melee)
- mineral gun
- helmet
- monsters (fly, crawl, fix spawn, boss)
- ambient object (critter, plant, background, vfx) non interactive
- crafted consumable (potion)
- crafted non-consumable (helmet, weapon, mineral gun, ladder, storage)
- placeable units (block, torch, ladder, craftable furniture/building)

Not visible

- target
- hitbox

Blocks

- 32 x 32 block size
- harvestable block = dirt, sand, copper, cobalt, ferrite, crystal silica, iron, gold, ammonia, cadmium, uranium, paraffin

Inventory

a 4x4 - 16 slots bag with option to 2 extra slot to craftable toolbelt

Boss fight

The first boss will spawn underground in an arena-like room which will require a special pickaxe to be breached. This arena will be randomly generated when the player's kill count reaches fifty.

The second boss room will spawn once the player crafts an item which is used to summon it. This summoned boss will be a flying creature (possibly spawn at the altar-type which is a craftable item).

Summoning boss

There is no plan to repeatedly summon bosses, since bosses are just part of the story progression and the item gained has a fixed chance (this could change in the future if we decide to lower the drop chance of the materials which is a stretched goal).

1st boss - tusk worm (could also drop ivory, can be used to extract purified keratin), think of the mechanical worm boss in Terraria

2nd boss - bat-like creature (could also drop meat that can be used as an antidote against the last boss)

3rd boss - giant broodmother (initiate ending and return home)

stage 1 - avoid projectile for 5 rounds (cannot fight back)

stage 2 - fight with its younglings (spit younglings on the ground and fight)

stage 3 - try to bite then fangs stuck for 3 sec (must avoid the bite then attack the head)

Project plan

week1

- UML and plans
- complete first version of game design doc

- draft art asset
- set up pipeline and meeting plans

week2

- character motion with asdf, jump, LM click to shoot/break a block
- generate terrain (to check size and memory issues, no cave needed)
- design game material tech tree
- flowchart for the whole project
- 1st draft UI (game menu, inventory, quick item, shop ui, crafting menu ui)

week3

- design weapon tech tree
- inventory and item UI interaction
- breakable block, placeable block
- first draft monsters, attack, locomotion, player hit feedback, death and spawn

week4

- hot bar (UI)
- two moving monsters with animation and AI
- terrain with caves and different brick types
- crafting mechanic, working craftable parts
- boss 1 fight design and 1st draft implementation
- **prototype** done for the game (character control, break block, place block, craft something that works as intended (maybe an antenna), bosses are not implemented at this point)

week5

- refine gameplay controller
- spawn all monsters
- lighting system
- all monsters implementation with all motion and projectiles
- health bar, armor stats, item durability implementation

week6

- crafting potions mechanics
- all plants are populated in the world and spawn correctly

week7

- boss mechanic finalized and working
- storytelling element and menu

week8

- buffer week for vfx and sound, title screen
- save/load game
- save point implemented

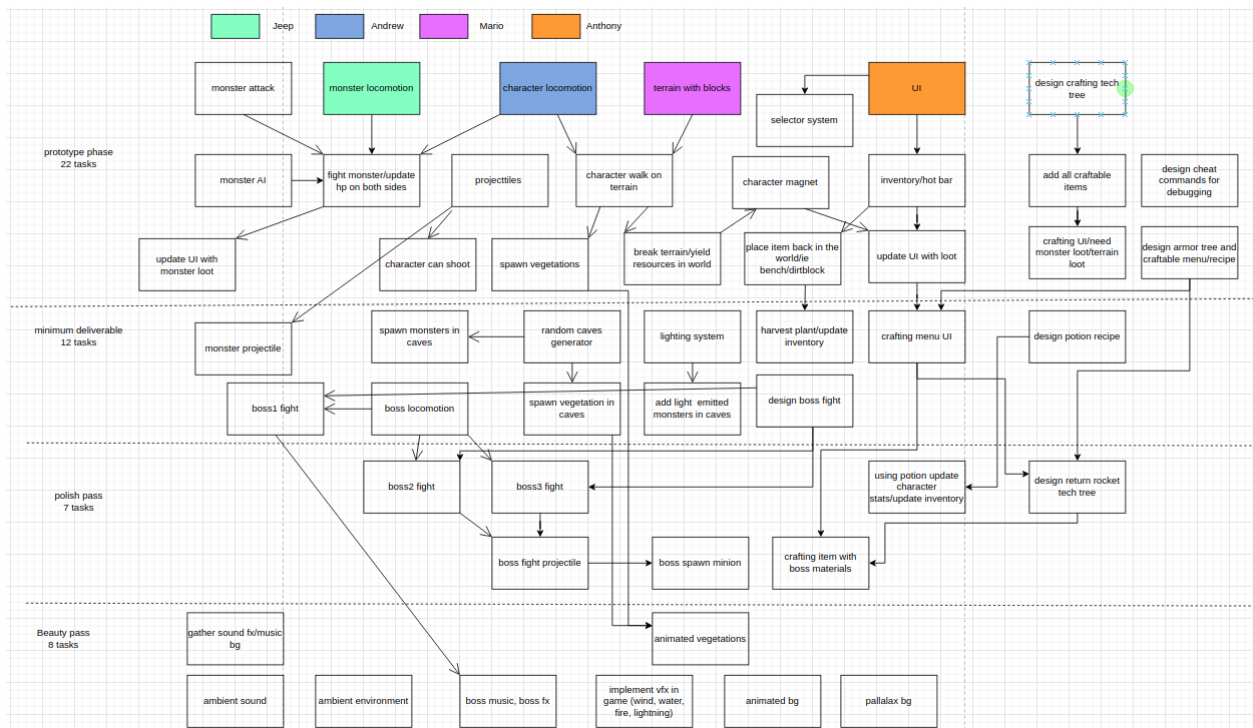
week9

- game tuning and buffer week

week10

- buffer for late stuff
- populate world - ambient animation

Game tech dependencies



Full Size